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Delta Hookup

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Outlines test procedures and methods for use in evaluating the requirements for electrical power. Includes checklist and data collection sheets.

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US ARMY TEST AND EVALUTATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-105
*Test Operations Procedure 6-2-514 15 October 1981
AD No. A106797

ELECTRICAL POWER REQUIREMENTS

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1. SCOPE

This procedure outlines the particular tests and test methods for use in evaluating the electrical power requirements of communications, surveillance, and electronic equipment and materiel. It serves as a guide when determining the overall efficiency of such equipment insofar as the electrical power requirements are concerned. Notwithstanding the accuracies, frequencies, and levels stated in this test operations procedure (TOP), the specific equipment requirements stated in the appropriate requirements documents must be used.

1.1 Subtests.

1.1.1 Warm-Up Power Requirements. The objective of the subtest is to determine the electrical power requirements of the test item during the warm-up period.

1.1.2 Range of Power Requirements. The objective of the subtest is to determine the power requirements of the test item and its power factor (if applicable) while at minimum load operation (standby) and at peak loading conditions.

1.1.3 Frequency Variation. The objective of the subtest is to measure the test item's power requirements during the presence of frequency variations in the primary power supply.

1.1.4 Voltage Variation. The objective of the subtest is to measure the effects of variations in the power supply voltage level on the test item's

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power requirements and to assess the effectiveness of the test item's low and high voltage protection devices.

1.2 Limitations:

1.2.1 Testing of power sources and power transmission lines is considered only with respect to their error contributions to the overall measurements. The demand factor of the power distribution system is not calculated in this test. Normally, the test item power requirements will be measured only at the point where the power transmission cable is mated to the test item. Available and reactive power are not primary considerations in this test procedure.

1.2.2 This test procedure does not address the spikes and transients induced upon the power distribution system by the test item.

2. FACILITIES AND INSTRUMENTATION

2.1 Facilities:

ITEM	REQUIREMENT
Floor and/or bench space	Under ambient laboratory conditions, adequate space for the equipment under test and the instrumentation to be used.
Electrical Power	Dependent upon the equipment undergoing test and the required instrumentation, but generally includes 117 VAC and 220 VAC, single and three-phase (Y and Delta); ± 5% of nominal rms volts.
AC Power Supply DC Power Supply	± 0.1%

2.2 Instrumentation:

ITEM	MINIMUM STANDARDS FOR ACCURACY/PURPOSE
Voltmeter	± 2.0 dB
Wattmeter	± 5% full scale reading
Frequency Meter	± 1% variation
Transformer	Used to step down voltages when it is anticipated that power consumed by test item is greater than wattmeter's capability.
Ammeter	Used to measure DC power and single phase AC power requirements. 0-60 AC Amps; DC to 400 HZ, ± 5% full scale reading.

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ITEM	MINIMUM ACCURACY/PURPOSE
Oscilloscope with Camera	Horizontal deflection = ± 3% Vertical deflection = ± 2%
Strip Chart Recorder	
Stop Watch	± 0.1% variation

3. PREPARATION FOR TEST

3.1 Planning. The test officer must assure himself that the test plan will sufficiently exercise the test item to accomplish the objectives for undertaking the test. For development tests (DT), the Independent Evaluation Plan (IEP) and the Test Design Plan (TDP) will usually outline the particular requirements to be included in the test plan. The assigned test officer will activate a project notebook for each test item, recording in it pertinent descriptive and technical information. The project notebook provides a narrative discussion of test conduct and results and is kept current for the duration of the test program. In addition, test planning encompasses a consideration of the potential foreign threat factors that will permit a realistic evaluation of the test item in a threat environment. Complete test planning requires that the test office:

a. Prepare a test operations checklist using appendix A as a guide.

b. Incorporate complete safety aspects within the preparations for the test.

c. Brief test personnel on all aspects of the test program to include the purpose of the test and the precision requirements during test conduct.

d. Provide sufficient copies of operating instructions to all test personnel.

3.2 Facilities. Preparation of the test facilities to be used for the conduct of electrical power requirements testing should include:

a. Adequate lead time for scheduling the test facilities that are to be used.

b. Preparation of laboratory benches and floor space.

3.3 Test Item. In his preparations for test conduct, the test officer will insure that:

a. A record of the test item's nomenclature, its technical characteristics, its manufacturer, and its performance parameters are entered into the project notebook.

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b. The test item is photographed from several perspectives to provide a nonambiguous visual identification and description of the test item.

c. The test item and all of its associated components are inspected for damage, deterioration, and obvious manufacturing defects.

d. The test item is in proper operating condition.

3.4 Instrumentation. In his preparations for test conduct, the test officer will insure that:

a. All test equipment and other instrumentation is scheduled and readily available for use during test conduct.

b. All test equipment is calibrated prior to the scheduled test start date and that calibration standards equal or exceed the standards stipulated in paragraph 2.2 above.

3.5 Equipment Setup: Prior to test conduct, the test officer must establish a plane of reference that clearly separates test measurements performed on the test item from the source of power.

3.5.1 Figure 1 shows the plane of reference with respect to the power source and the test item (or power sink) and the intervening positions of the respective items of instrumentation.

3.5.2 Figure 2 shows the instrumentation setup for direct current (DC) power measurements.

3.5.3 Figure 3 shows instrumentation setups for three acceptable methods of taking single-phase alternating current (AC) power measurements.

3.5.4 Figure 4 shows the instrumentation setup for three-phase, three-wire AC power requirements.

3.5.5 Figure 5 shows the instrumentation setup for four-wire AC power measurements.

4. TEST CONTROLS

4.1 Facilities:

a. The test facility's ambient temperature should not vary more than 15°C.

b. Line voltages used to provide a power source to the test item should not vary more than five percent from the mean.

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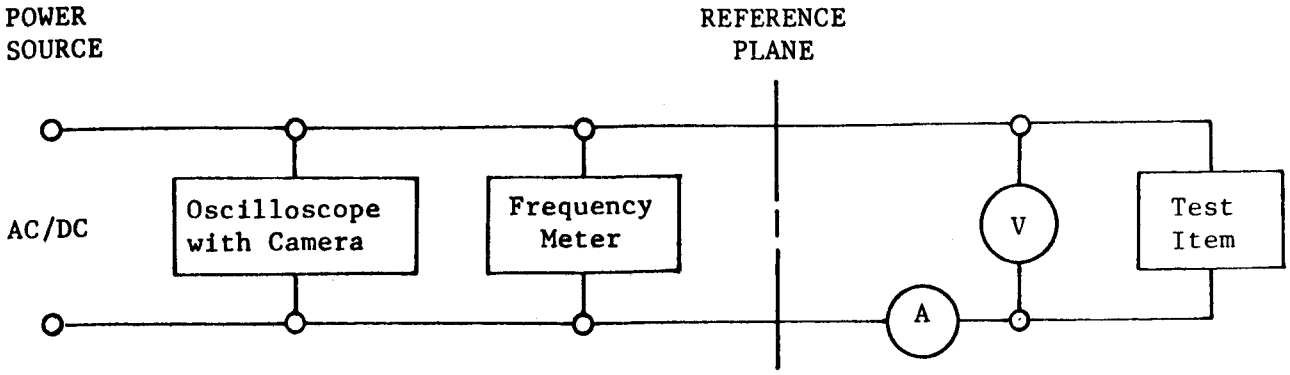


Figure 1 - Plane of Reference and Instrumentation Relationships

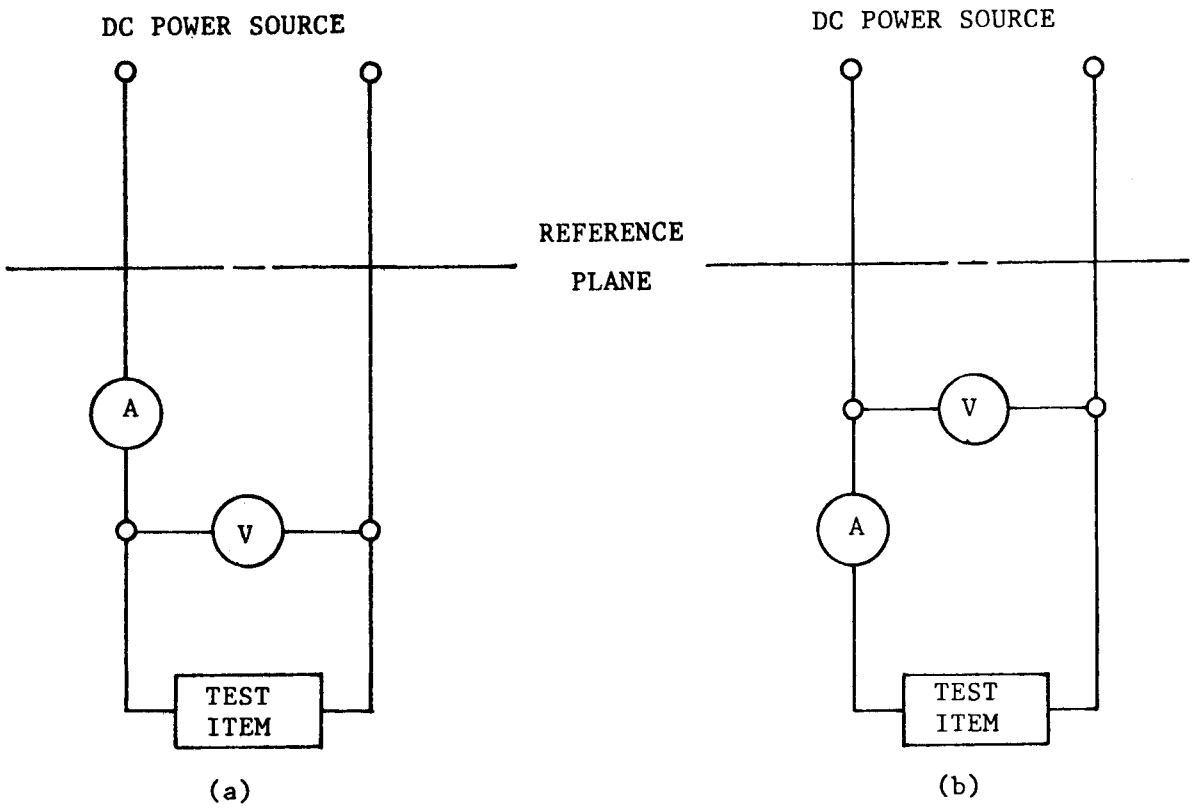


Figure 2 - DC Power Measurements Using Voltmeter/Ammeter Method

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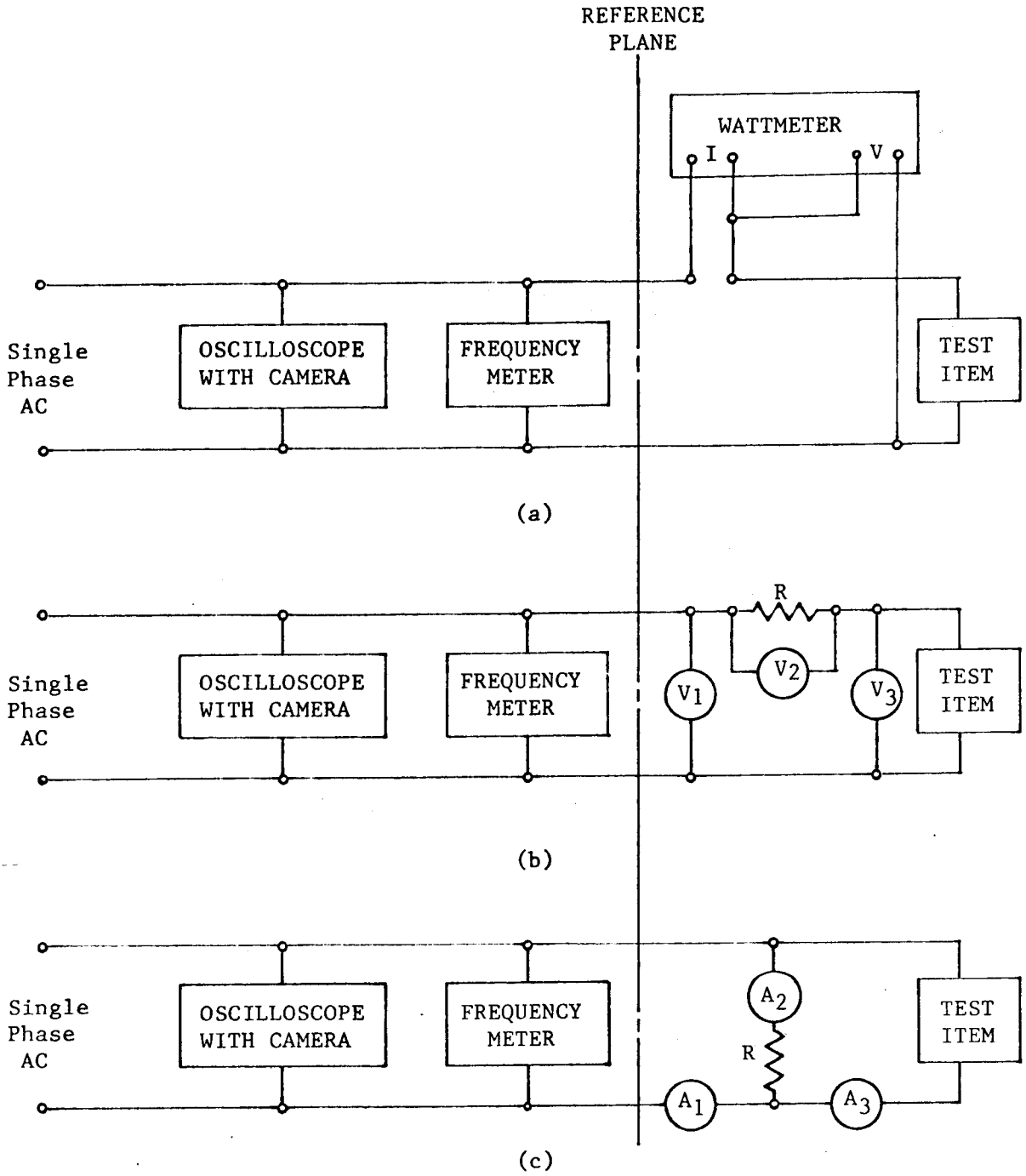


Figure 3 - Equipment Setups for Monitoring Single Phase AC Power Requirements Using (a) Wattmeter Method; (b) Three-Voltmeter Method; or (c) Three-Ammeter Method

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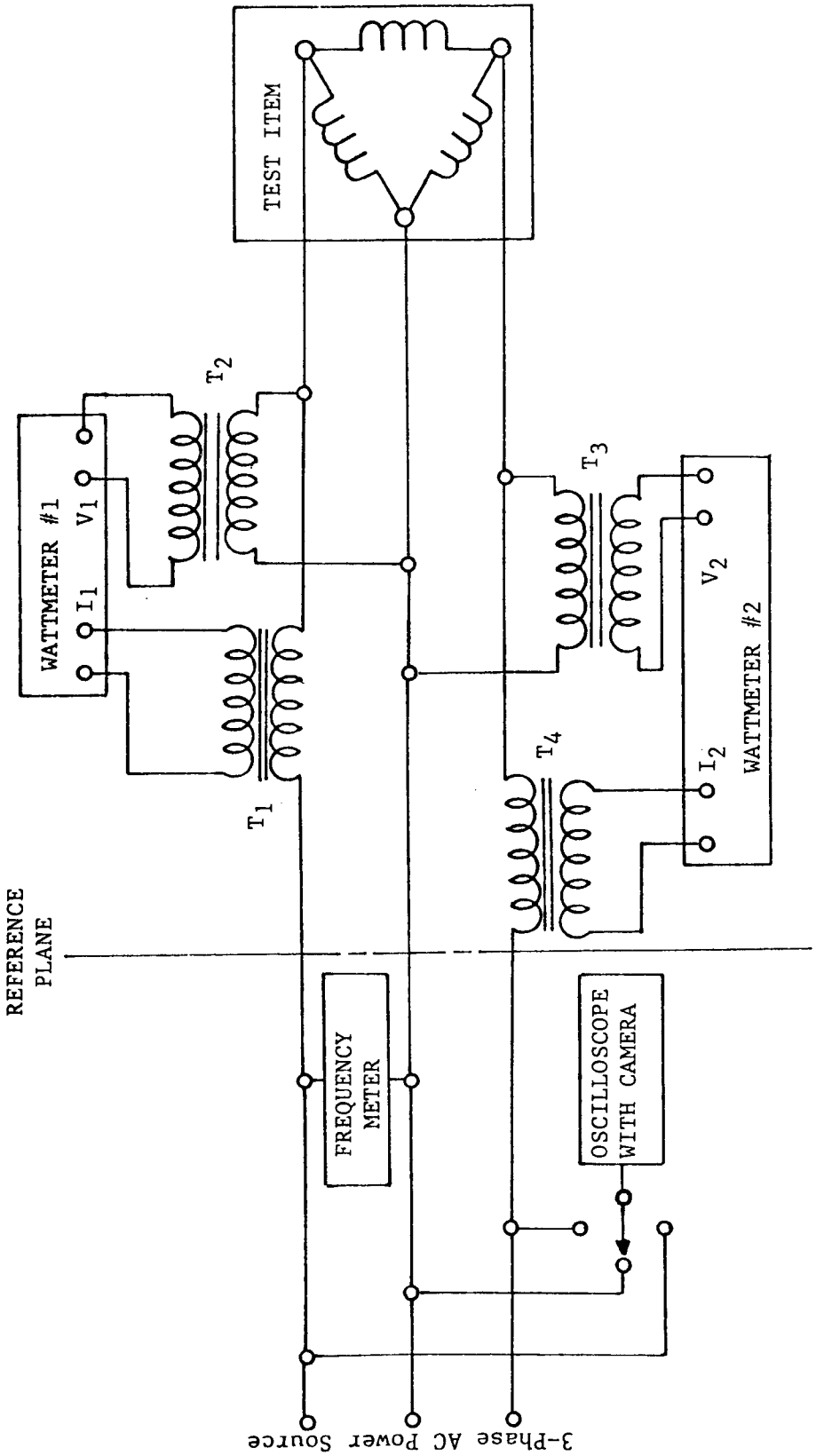


Figure 4 - Equipment Setup to Measure Three-Phase Power Requirements Using Two Wattmeters

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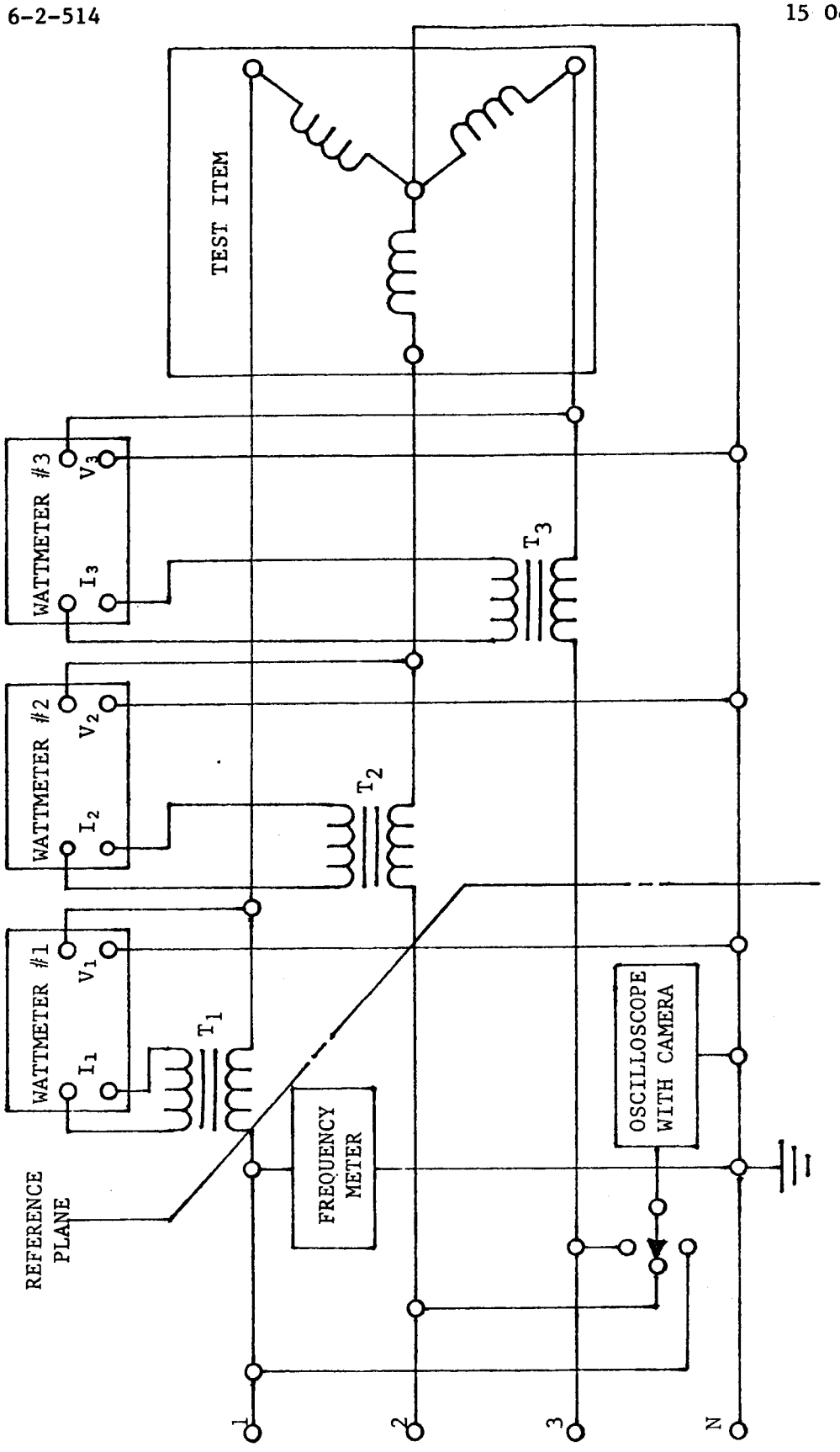


Figure 5 - Equipment Setup to Measure Four-Wire AC Power Requirements Using Three Wattmeters

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c. Laboratory power supplies will be tested prior to test start to catalog their electrical characteristics. A DC power supply will have an rms (root mean square) pard (periodic and random deviation) no greater than 1% and an AC power supply will have a harmonic distortion no greater than 0.5%.

4.2 Test Item:

a. Select a convenient, recordable, reference level for use with the test item, e.g., transmitter output, receiver sensitivity, frequency response, or dynamic gain, to monitor test item performance in the presence of input power and/or frequency variations. Operate the test item to draw rated power and record the level of the operating parameter as a standard bench mark for future reference.

b. Except as directed by test procedures, the test item will not be moved, adjusted, or calibrated while electrical power tests are in progress.

4.3 Instrumentation:

a. Applicable instrumentation will satisfy the standards for accuracy and stability outlined in paragraph 2.2 above.

b. Instrumentation will be calibrated prior to conduct of the test.

c. The same instrumentation will be used throughout a specific sub-test.

d. A step-down transformer will be inserted into the test circuit in a manner similar to figure 4 or 5 when it can be anticipated that the power consumed by the test item will be greater than the capacity of the wattmeter.

5. PERFORMANCE TESTS

5.1 Method.

5.1.1 Warm-Up Power Requirements:

a. Insure that the test item is turned off and has been turned off long enough to reach the ambient temperature.

b. Connect the test item to a power source that can supply the nominal voltage and frequency, when applicable, stated in the manufacturer's specifications.

c. Connect the test instrumentation as depicted in figures 2 through 5, using the equipment setup that is appropriate for the type of power source and equipment under test.

d. Turn the test item on and operate it in a manner that requires instant rated power from the power source.

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e. Record data at 10-second intervals[1] from all instrumentation until a steady state condition is achieved or the standard reference level selected in paragraph 4.2a above is reached, whichever takes longer.

5.1.2 Range of Power Requirements:

a. Connect the test item to a power source capable of supplying the nominal voltage and frequency, when applicable, stated in the manufacturer's specifications.

b. Connect the test instrumentation as depicted in figures 2 through 5, using the equipment setup that is appropriate for the type of power source and equipment under test.

c. Allow sufficient time for test item warm-up and adjust the test item's controls to a point where it will consume minimum power.

d. Activate all recording instrumentation.

e. Adjust the test item's control settings so as to increase the power supply loading enough to develop a trend and record the control setting position and instrumentation readings.

f. Repeat step e, enough times to produce a smooth curve, until the test item is operating at peak load conditions. A minimum of four different control settings should be used.

g. If the test item contains environmental control equipment that is used to cool or heat the assemblage, this subtest must be repeated at the environmental temperature extremes specified for the test item.

5.1.3 Frequency Variation:

a. Connect the test item to a variable frequency power supply adjusted to deliver the nominal voltage and frequency stated in the manufacturer's specifications.

b. Connect the test instrumentation as depicted in figures 3 through 5, using the equipment setup that is appropriate for the type of power source and equipment under test.

c. Maintain a constant input voltage level while varying the power supply frequency in incremental steps as follows:

(1) From 55 to 65 Hz for 60 Hz power requirements.

(2) From 380 to 420 Hz for 400 Hz power requirements.

1/ Data may be recorded continuously by connecting the instrumentation to a strip chart recorder, when using appropriate test instrumentation, e.g., HP-7562 or HP-3400 voltmeters.

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d. Record the standard reference indicator levels and control settings selected in paragraph 4.2a above for a sufficient number of frequencies to enable plotting a smooth frequency curve as a function of the standard reference indicator levels.

5.1.4 Voltage Variations (Overvoltage and Undervoltage):

a. Connect the test item to a variable voltage power supply adjusted to deliver the nominal rated voltage and frequency, when applicable, stated in the manufacturer's specifications.

b. Connect a voltmeter across the power source side of the reference plane.

c. Connect monitoring instrumentation as depicted in figures 2 through 5, using the equipment setup that is appropriate for the type of power source required.

d. Allow sufficient time for test item warm-up; then decrease the power supply voltage in small decrements.

e. Monitor and record the corresponding power levels and the standard reference indicator levels and control setting selected in paragraph 4.2a above.

f. Stop testing when the test item's undervoltage protection devices operate.

g. From zero volts, increase the power supply voltage. Record the voltage at which the test item's standard reference indicator initially responds and also when it finally achieves the standard reference level.

h. Increase the power supply voltage from its nominal rated voltage in incremental steps.

i. Monitor and record the corresponding power levels, the standard reference indicator levels, and control setting selected in paragraph 4.2a above.

j. Stop testing when the test item's overvoltage protection devices commence to operate.

5.2 Data Required.

5.2.1 Warm-Up Power Data Requirements:

a. The ambient temperature in degrees Celsius.

b. The input power in watts as a function of time.

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c. The input voltage waveform (if applicable).

d. When applicable, the frequency of the power source.

5.2.2 Range of Power Data Requirements:

a. The input power in watts as a function of the test item's control setting.

b. When applicable, the input frequency in hertz as a function of the test item's control setting.

c. The input voltage waveform as a function of the test item's control setting.

5.2.3 Frequency Variation Data Requirements:

a. The standard reference indicator level as a function of input frequency.

b. The input power in watts as a function of frequency.

c. The input voltage waveform at each frequency.

d. The ambient temperature in degrees Celsius.

5.2.4 Voltage Variation Data Requirements:

a. The standard reference indicator level as a function of voltage.

b. The input power in watts as a function of input voltage.

c. The input voltage at which the test item's undervoltage protection device activates (or the test item fails).

d. The input voltage at which the test item's overvoltage protection device activates (or the test item fails).

e. The input voltage waveform and frequency.

f. The ambient temperature in degrees Celsius.

6. DATA REDUCTION AND PRESENTATION

6.1 Data Reduction. For test items that are likely to consume large amounts of power, it may be necessary to insert a step-down transformer into the circuit. When this situation arises, use the manufacturer's specifications for the transformer to convert the wattmeter readings into true wattage for data reduction measurements. Four types of power sources can be used and the instrumentation configurations for each are shown in figures 2 through 5. The following procedures apply depending on the type of power source to be employed:

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6.1.1 Direct Current (DC)

a. Power measurements will require correction for the power dissipated in the ammeter and voltmeter used.

b. Calculate load power using the appropriate formulas below, depending on whether the voltmeter resistance, R_v, or the ammeter resistance R_a, is known:

$$P = EI - \frac{E^2}{R_v}$$ (see figure 2a, page 5)

$$P = EI - I^2R_a$$ (see figure 2b, page 5)

6.1.2 Single Phase Alternating Current (AC):

a. Power requirements may be measured directly using the wattmeter method (figure 3a); however, the wattmeter method cannot be used to determine the power factor.

b. Power in the load can be determined using the three-voltmeter method (figure 3b) by substituting the respective voltmeter readings and the known resistance into the following formula:

$$P = \frac{(V_1)^2 - (V_2)^2 - (V_3)^2}{2R}$$

If the current in the resistance, R, is the same as the load current and the resistance is noninductive, the power factor, cos θ, can be calculated from the values used to determine the power in the load:

$$\cos \theta = \frac{(V_1)^2 - (V_2)^2 - (V_3)^2}{2V_2V_3}$$

c. Power in the load can also be determined using the three-ammeter method (figure 3c) by substituting the respective ammeter readings and the known resistance into the following formula:

$$P = \frac{(I_1^2 - I_2^2 - I_3^2)R}{2}$$

The power factor may also be calculated from the ammeter values used to compute the power in the load with the formula shown below:

$$\cos \theta = \frac{(I_1^2 - I_2^2 - I_3^2)}{2I_2I_3}$$

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6.1.3 Three-Phase, Three-Wire (Delta-Load) AC:

a. Normally, the two-wattmeter method (figure 4) will be used to measure three-phase, three-wire power requirements of the test item.

b. Let W_1 and W_2 represent the respective wattmeter readings. $W_1 + W_2$ equals the total power required by the test item.

c. Use the formula shown below to calculate tan θ:

$$\tan \theta = \frac{\sqrt{3}(W_2 - W_1)}{W_1 + W_2}$$

d. From tan θ determine the value of θ and the power factor, cos θ, can be derived.

6.1.4 Three-Phase, Four-Wire (Y of Star-Load) AC: AC power measurements are taken using the three-wattmeter method shown in figure 5. The sum of the wattmeter readings, $W_1 + W_2 + W_3$ equals the total power required by the test item. The power factor can be determined using the same formula as in paragraph 6.1.3c.

6.2 Data Presentation.

6.2.1 Warm-Up Power Requirements:

a. In a format similar to that shown in appendix B, page B-3, tabulate warm-up data.

b. Present warm-up power data separately for each ambient temperature at which measurements were taken.

c. Present data graphically to show input power in watts as a function of time in seconds; and the frequency of the power source in hertz as a function of time in seconds.

6.2.2 Range of Power Requirements:

a. In a format similar to that shown in appendix B, page B-4, tabulate range of power data.

b. Present data graphically to show input power in watts as a function of the control setting; and the frequency of the power supply in hertz as a function of the control setting.

6.2.3 Frequency Variation Power Requirements:

a. In a format similar to that shown in appendix B, page B-5, tabulate frequency variation data.

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b. Present data graphically to show the standard reference levels as a function of the power supply's input frequency; and the input power in watts as a function of the input frequency.

6.2.4 Voltage Variation Requirements:

a. In a format similar to that shown in Appendix B, page B-6, tabulate voltage variation data, from undervoltage protection device activation to overvoltage protection device activation (or equipment failure).

b. Present voltage variation data separately for each ambient temperature at which measurements were taken.

c. Present data graphically to show the standard reference indicator levels as a function of input voltage; and input power in watts as a function of input voltage.

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APPENDIX A

CHECKLIST FOR ELECTRICAL POWER REQUIREMENTS TEST

	INITIALS	
	Test Officer	Test Supervisor
Availability of appropriate documentation and authority for the conduct of the test program:	_____	_____
Initiate a project notebook:	_____	_____
Establish the project file:	_____	_____
Availability of appropriate references:		
Required Operational Capability (ROC):	_____	_____
Test Item Specifications:	_____	_____
Military Standards:	_____	_____
Test Operations Procedures:	_____	_____
Operating Manuals:		
Coordination with test sponsor/contractor:	_____	_____
Are dates for desired test execution realistic?	_____	_____
Availability of test plan to all participants:	_____	_____
Coordination for necessary test support:	_____	_____
Identification of critical test issues:	_____	_____
Test facility conforms to specified standards:	_____	_____
Brief all test personnel:	_____	_____
Instrumentation data have been recorded:	_____	_____
Instrumentation has been currently calibrated:	_____	_____
Calibration due date for all instrumentation will permit completion of testing:	_____	_____
Test item data recorded in project notebook:	_____	_____
Initial inspection performed:	_____	_____
Photographs taken of test item(s):	_____	_____
Equipment performance reports prepared:	_____	_____
Data collection sheets are complete:	_____	_____
Test report prepared:	_____	_____

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APPENDIX B

DATA COLLECTION SHEETS AND PRESENTATION FORMATS

Sheet #1	Electrical Power Requirements Data Sheet	B-2
Sheet #2	Warm-Up Power Requirements	B-3
Sheet #3	Range of Power Requirements	B-4
Sheet #4	Frequency Variation Power Requirements	B-5
Sheet #5	Voltage Variation Power Requirements	B-6

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ELECTRICAL POWER REQUIREMENTS DATA SHEET

Voltage: AC ☐ DC ☐ Frequency: _____Hz Phase: Single ☐ Triple ☐

Two-Wire ☐ Three-Wire ☐ Four-Wire ☐

Normal _____Volts Environmental Temperature _____°C

Trial Number	V_1	V_2	V_3	A_1	A_2	A_3	W_1	W_2	W_3	T_1	T_2	T_3	T_4

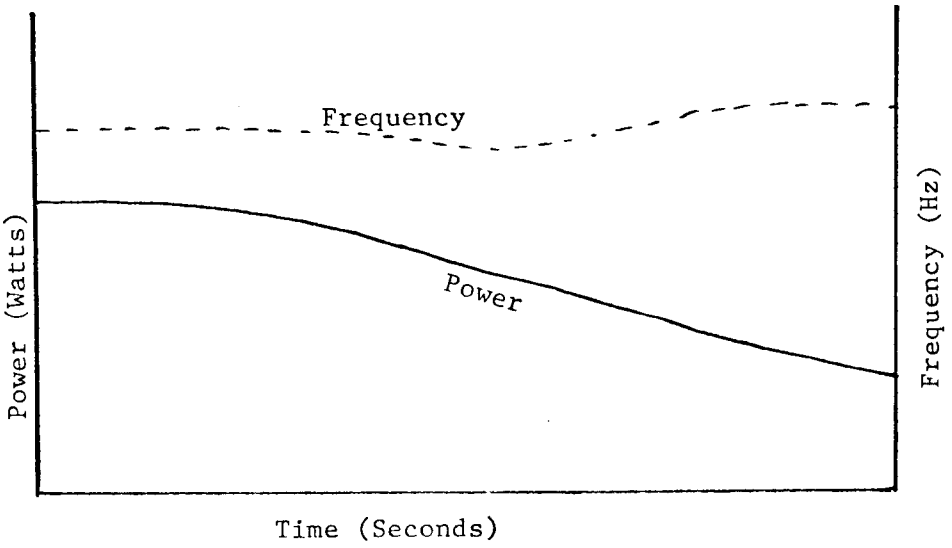
Use only those columns required by the test setup selected for each subtest.
Label columns to fit data requirements.

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SHEET #2

WARM-UP POWER REQUIREMENTS

Time	Power Requirement	Hertz



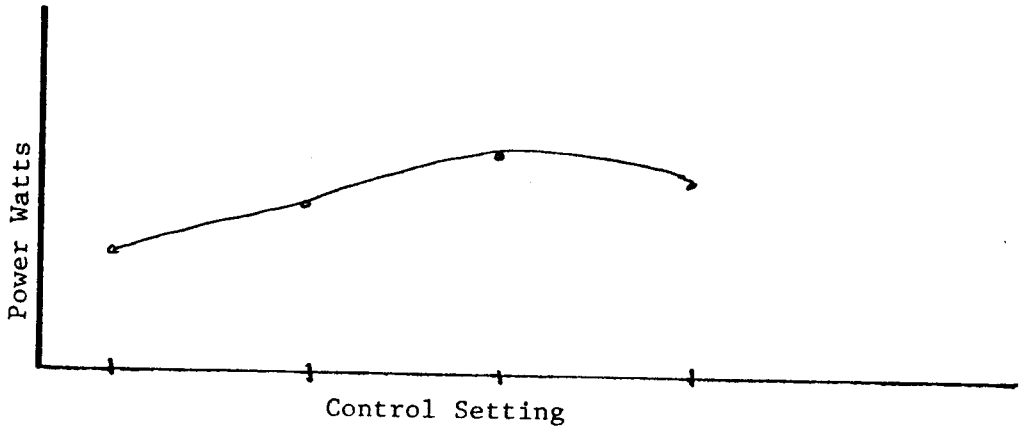
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SHEET #3

RANGE OF POWER REQUIREMENTS

Control Setting*	Power Required

*Identify which control is used to vary the power requirements.



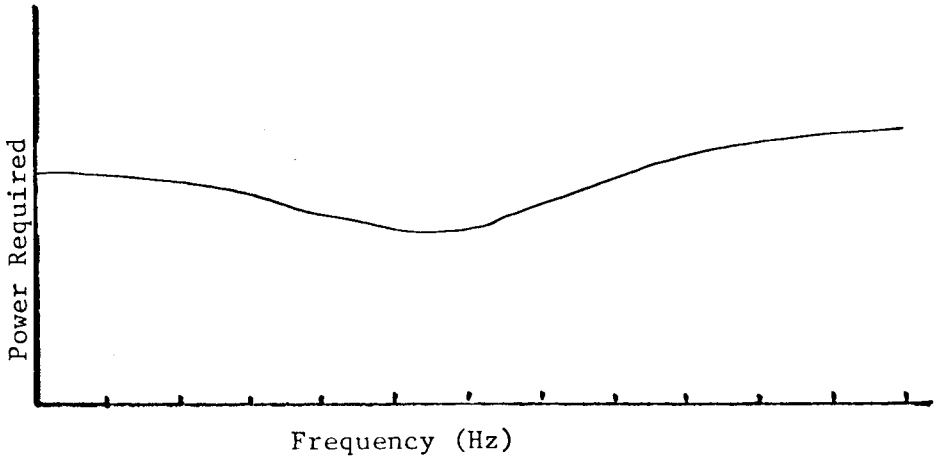
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SHEET #4

FREQUENCY VARIATION POWER REQUIREMENTS

Frequency (Hertz)	Power Required
*	

*Center frequency should be specification frequency.

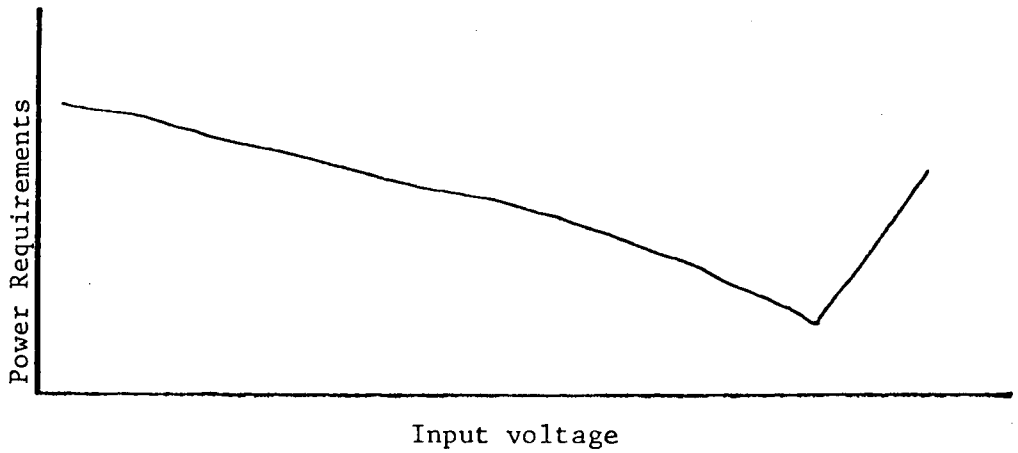


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SHEET #5

VOLTAGE VARIATION POWER REQUIREMENTS

Voltage	Power Required	Remarks
		Undervoltage Protection Device Activates
		Center Voltage (Specification Voltage)
		Overvoltage Protection Device Activates



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